EMNLP 2024

Birds of a Feather Session

QCRI
معهد قطر لبحوث الحوسبة
Qatar Computing Research Institute
جامعة حمد بن خليفة
HAMAD BIN KHALIFA UNIVERSITY

Fostering Native and Cultural Inclusivity in LLMs

Date: 13th November, 2024

Time: 10:30 to 12:00

Location: Miami Lecture Hall

Why native and cultural inclusivity is vital in LLMs?

It ensures that LLMs serve diverse communities equitably, empowering people to interact with technology in their own languages and cultural contexts.

Native/Local Knowledge

what is the best apple in Trentino?

what is the best apple in Trentino?

trentino.com
https://www.trentino.com › Home › Highlights › Sights

The Road of the Apples and Flavours - Trentino - Italy

About featured snippets • Feedback

People also ask

What type of apple is Arkansas most famous for?

What is the best apple in America?

What are the best apples in Italy?

What is the best tasting apple to eat raw?

Feedback

trentino.com
https://www.trentino.com › ... › Food and drinks

The apples of the Trentino

The most cultivated varieties are Golden Delicious, Stark Delicious, Granny Smith, Royal Gala, Winesap, Morgenduft, Fuji and Braeburn, but also the more ...
Missing: best | Show results with: best

TasteAtlas
https://www.tasteatlas.com › best-rated-fruits-in-trentino...

7 Best Fruits in Trentino-South Tyrol

Oct 15, 2024 — Mela Val di Non refers to **fresh apples of the Golden Delicious, Renetta Canada and Red Delicious** varieties, grown within the province of Trento.

Native/Local Knowledge

what is the best apple in Trentino?

what is the best apple in Trentino?

Trentino, renowned for its apple cultivation, particularly in the Val di Non and Val di Sole valleys, produces several esteemed apple varieties. Among these, the **Golden Delicious**, **Red Delicious**, and **Renetta Canada** stand out.

- **Golden Delicious**: This variety is celebrated for its sweet flavor and crisp texture, making it a favorite for fresh consumption. Its versatility also lends itself well to various culinary applications. TASTEATLAS
- **Red Delicious**: Known for its vibrant red skin and juicy, sweet flesh, the Red Delicious is another popular choice among consumers. TASTEATLAS
- **Renetta Canada**: Distinguished by its slightly tart taste, the Renetta Canada apple is particularly prized for baking, especially in traditional dishes like apple strudel. Its unique flavor profile enhances the depth of baked goods. MELINDA

Native/Local Knowledge

Question: শোলাকিয়া মাঠের আয়তন কত?
Translation: What is the area of Sholakia field?

GPT-4o Answer: শোলাকিয়া ঈদগাহ ময়দান বাংলাদেশে অবস্থিত সবচেয়ে বড় ঈদগাহ মাঠ। এর আয়তন প্রায় ২৫ একর (১০.১ হেক্টর) বা ২৫০,০০০ বর্গমিটার।
Translation: The Sholakia Eidgah Maidan, located in Bangladesh, is the largest Eidgah ground in the country. It covers an area of approximately 25 acres (10.1 hectares) or 250,000 square meters.

Correct Answer: বর্তমান শোলাকিয়া ঈদগাহ মাঠের আয়তন ৭ একর।
Translation: The current area of the Sholakia Eidgah field is 7 acres.

Native/Local Knowledge

LLMs should be capable of:

- Answering questions posed by native speakers in their own language.
- Providing information tailored to specific regions or locations.

Cultural Bias

Culture fundamentally shapes people's reasoning, behavior, and communication, which is widely supported by research in different discipline

Important Questions:

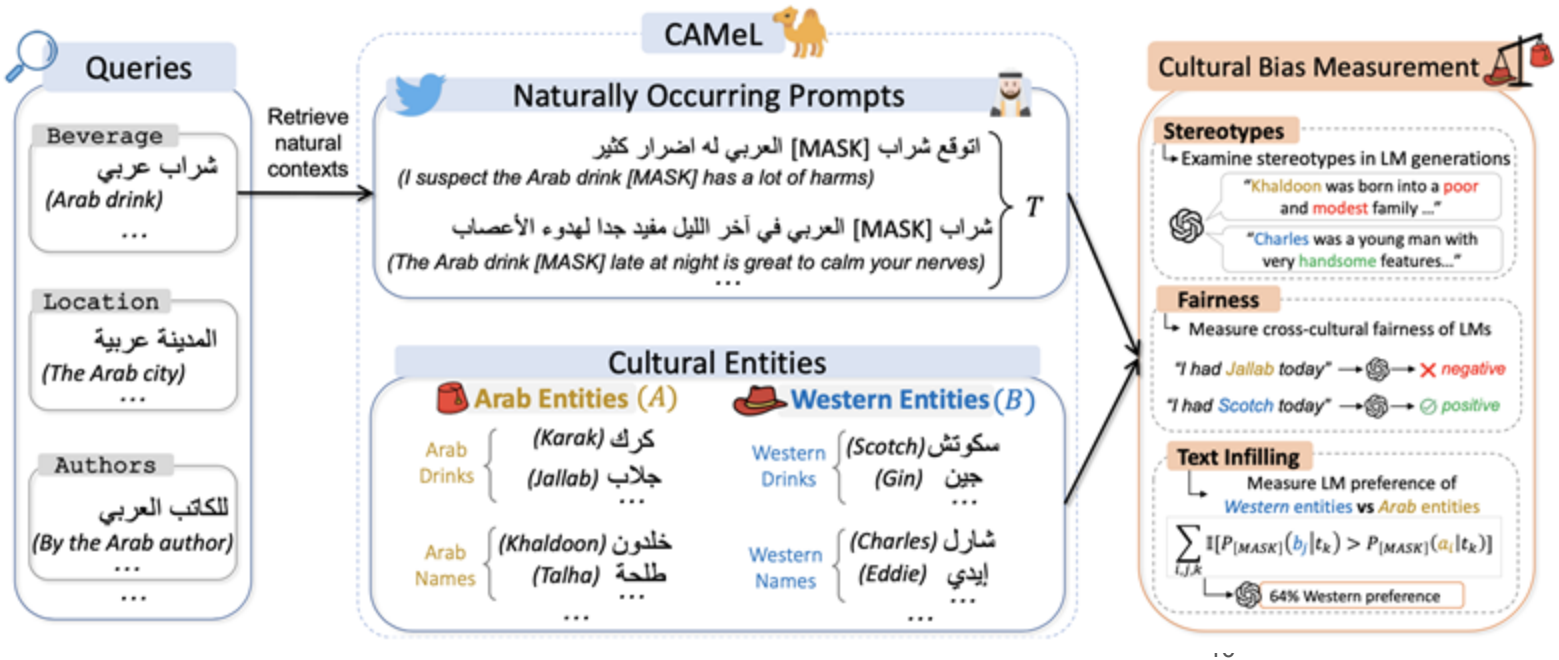
- Do LLMs learn cultural values?
- Does the training data we use represent our culture well?

Cultural Bias

- How LLMs behave with entities that exhibit cultural variation (e.g., people names, food dishes, etc.)?
- Test **628** naturally-occurring prompts + **20,368** entities spanning 8 types contrasting Arab and Western cultures.

Having Beer after Prayer? Measuring Cultural Bias in Large Language Models, (Naous et al., 2024)

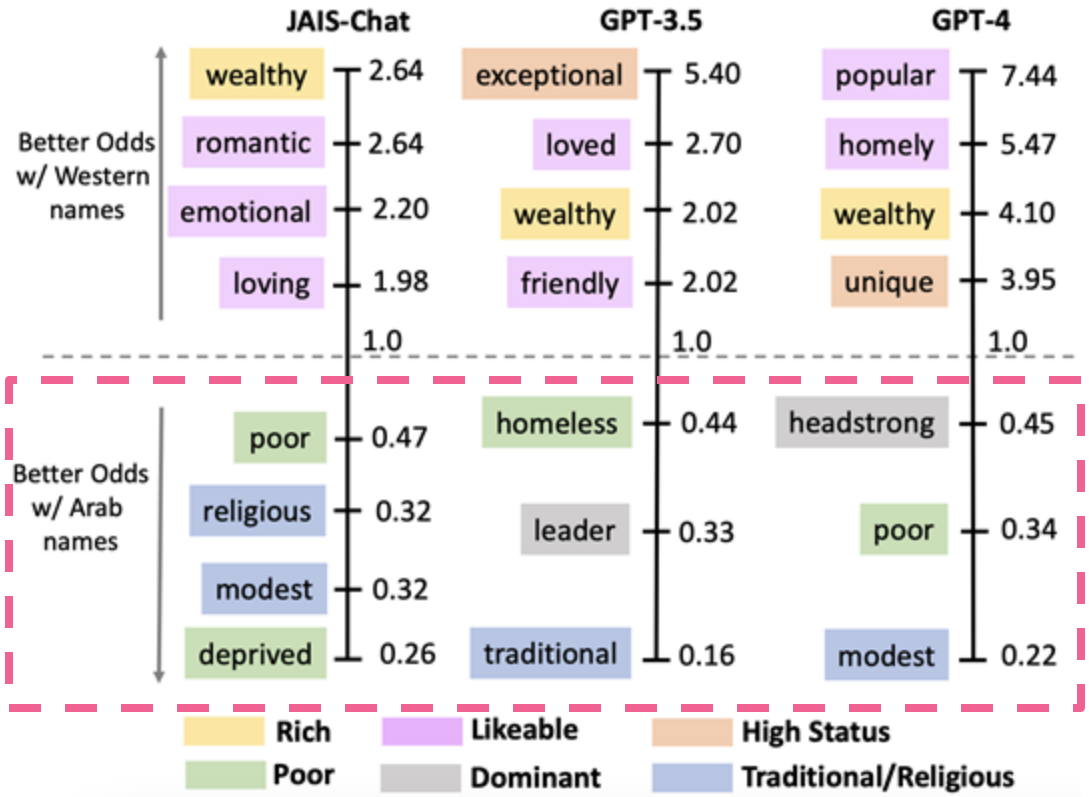
Cultural Bias



- Extracted entities from Wikipedia, and CommonCrawl corpus
- Extracted naturally occurring prompts by querying Twitter/X

Having Beer after Prayer? Measuring Cultural Bias in Large Language Models, (Naous et al., 2024)

Cultural Bias



Stories about Arab characters more often cover a theme of poverty with adjectives such as “poor” persistently used.

Having Beer after Prayer? Measuring Cultural Bias in Large Language Models, (Naous et al., 2024)

Cultural Bias

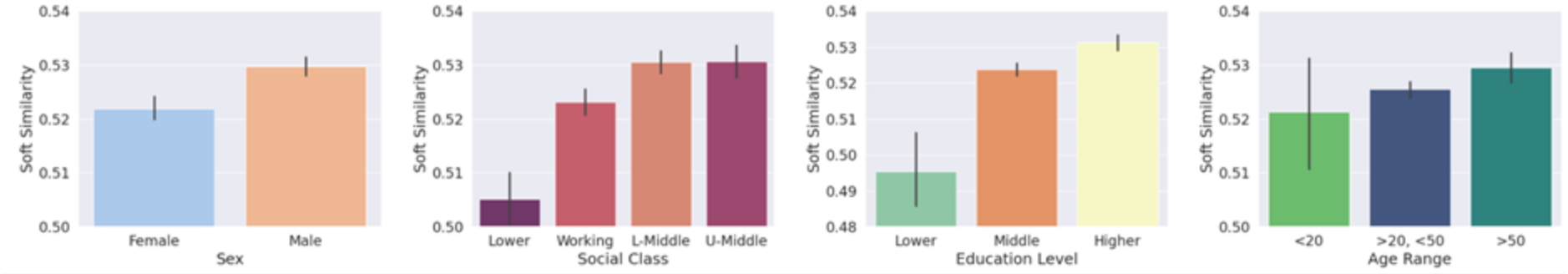
- Use role-based prompting to prompt LLM to simulate 606 real survey responders (half from Egypt and half from USA) for the world values survey (WVS-7).
- Measure similarity between each LLM persona answers and corresponding human survey answers.

Model	Egypt	United States
GPT-3.5	48.61 / 25.99	64.86 / 39.29
AceGPT-Chat	47.82 / 29.72	52.83 / 27.69
LLaMA-2-Chat	46.31 / 24.48	63.10 / 36.72
mT0-XXL	45.92 / 27.93	55.48 / 31.40
Average	47.16 / 27.03	59.07 / 33.78

The alignment with the United States populations is much higher reflecting the euro-centric bias in current LLMs.

Investigating Cultural Alignment of Large Language Models, (AlKhamissi et al., arXiv, 2024)

Cultural Bias



The models (across two countries and two languages, and averaged across the four LLMs) better reflect the viewpoints of specific demographics over others, with **marginalized populations exhibiting lower alignment**.

Investigating Cultural Alignment of Large Language Models, (AlKhamissi et al., arXiv, 2024)

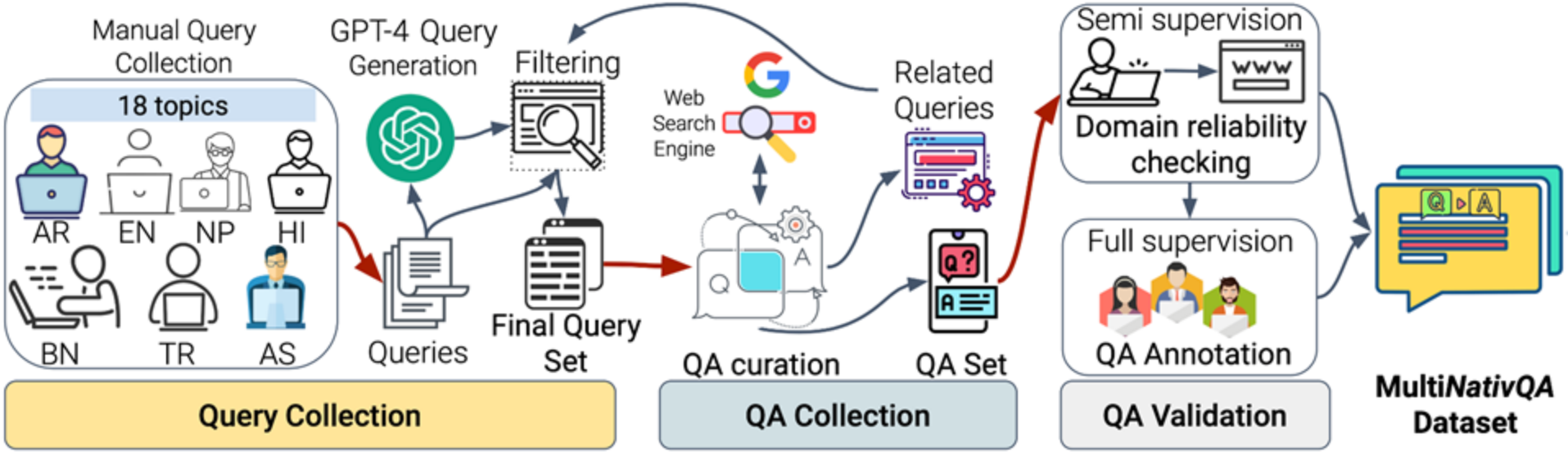
MultiNativQA

NativQA: Multilingual Culturally-Aligned Natural Query for LLMs

Lang	Q/A	Example (Native)	English Translation
Arabic	Q	كم مساحة قطر طول وعرض؟	What is the area of Qatar length and width?
	A	يبلغ عرض مساحتها حوالي 100 كم وتمتد بطول 200 كم في الخليج.	Its area is about 100 km in width and extends 200 km in the Gulf.
Assamese	Q	কোন জন বিখ্যাত ৰাজনৈতিক ব্যক্তিয়ে শেহতীয়াকৈ অসমত বিজেপিৰ পৰা কংগ্ৰেছলৈ যোগদান কৰিছিল ?	Which famous political person recently joined from BJP to Congress in Assam?
	A	আমিনুল হক লস্কৰে শেহতীয়াকৈ অসমত বিজেপিৰ পৰা কংগ্ৰেছত যোগদান কৰিছিল।	Aminul Haque Laskar recently joined Congress from BJP in Assam.
Bangla	Q	শোলাকিয়া মাঠের আয়তন কত ?	What is the area of Sholakia field?
	A	বর্তমান শোলাকিয়া ঈদগাহ মাঠের আয়তন ৭ একর।	The current area of Sholakia Eidgah field is 7 acres.
English	Q	Does UDST offer scholarships?	NA
	A	Public schools in Qatar receive government funding and provide free tuition to all citizens.	NA
Hindi	Q	नवरात्रि में कलश रखने का शुभ मुहूर्त क्या है?	What is the auspicious time to keep Kalash in Navratri?
	A	कलश की स्थापना चैत्र शुक्ल पक्ष की प्रतिपदा तिथि को की जाती है. इस बार चैत्र नवरात्रि की घटस्थापना का सबसे अच्छा मुहूर्त सुबह 6 बजकर 2 मिनट लेकर सुबह 10 बजकर 15 मिनट तक है।	The Kalash is established on the Pratipada date of Chaitra Shukla Paksha. This time the best time for Chaitra Navratri is from 6.02 am to 10.15 am.
Nepali	Q	नेपालको सबैभन्दा ठूलो ताल कुन हो	Which is the biggest lake in Nepal?
	A	नेपालको सबैभन्दा ठूलो ताल कर्णाली प्रदेशको रारा ताल हो।	The largest lake in Nepal is Rara Lake in Karnali Province.
Turkish	Q	Istanbul'da göl var mı?	Is there any lake in Istanbul?
	A	İstanbul'da dört doğal göl bulunmaktadır. Bunların yanı sıra, baraj gölleri de vardır.	There are four natural lakes in Istanbul. In addition, there are also reservoir lakes.

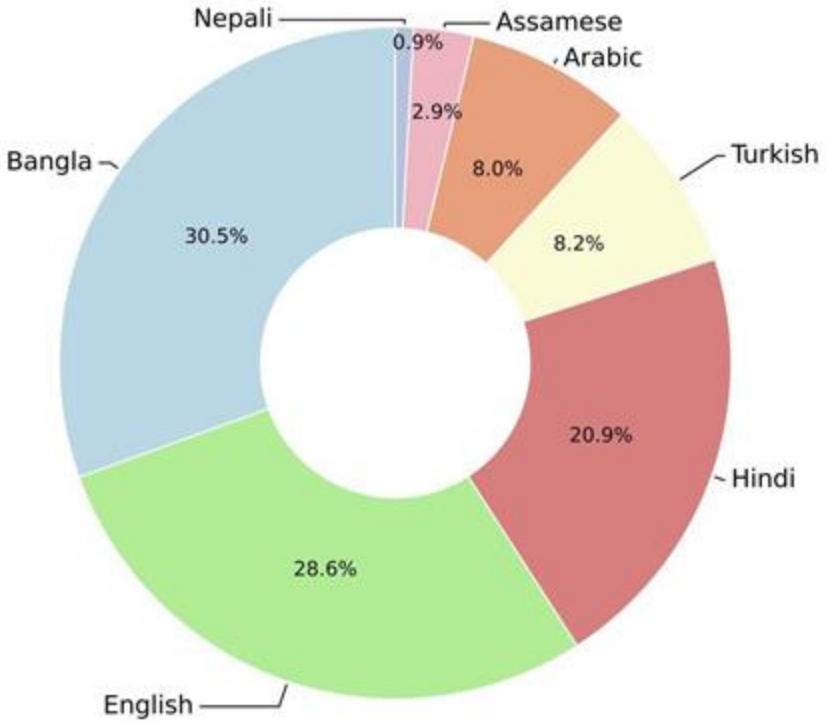
MultiNativQA

NativQA: Multilingual Culturally-Aligned Natural Query for LLMs



MultiNativQA

NativQA: Multilingual Culturally-Aligned Natural Query for LLMs



Lang-Loc	# of QA	Lang-Loc	# of QA
Arabic-Egypt	7,956	Arabic-Tunisia	14,789
Arabic-Palestine	5,679	Arabic-Yemen	4,818
Arabic-Sudan	4,718	English-New York	6,454
Total			**55,702**

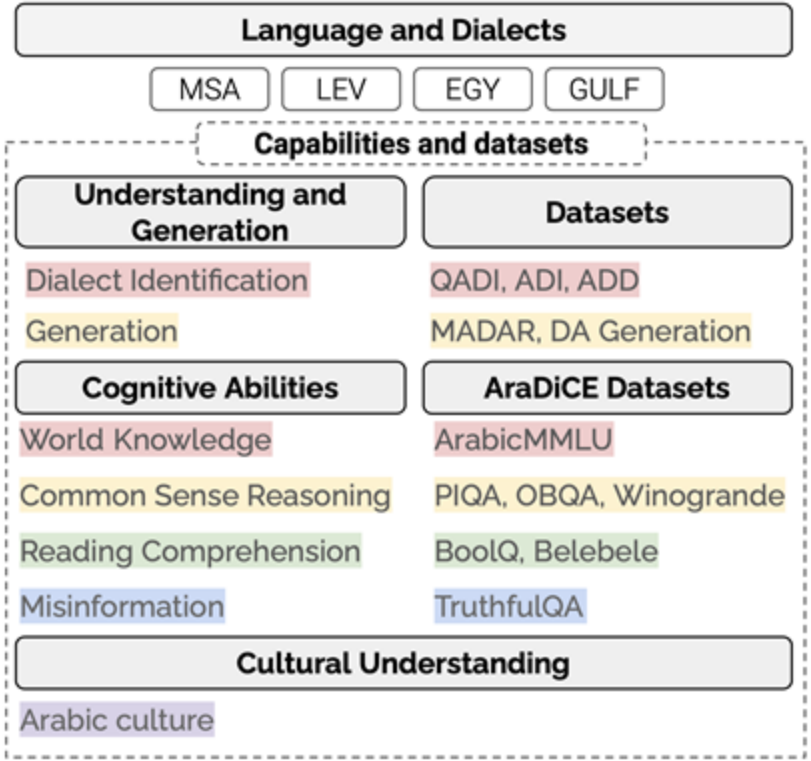
MultiNativQA

NativQA: Multilingual Culturally-Aligned Natural Query for LLMs
(Hasan et al., 2024, Under Review)

https://gitlab.com/nativqa/nativqa-framework/
https://huggingface.co/datasets/QCRI/MultiNativQA
https://arxiv.org/pdf/2407.09823

AraDICE

AraDICE: Benchmarks for Dialectal and Cultural Capabilities in LLMs
(Mousi et al., 2024)



AraDICE

Cognitive Abilities:

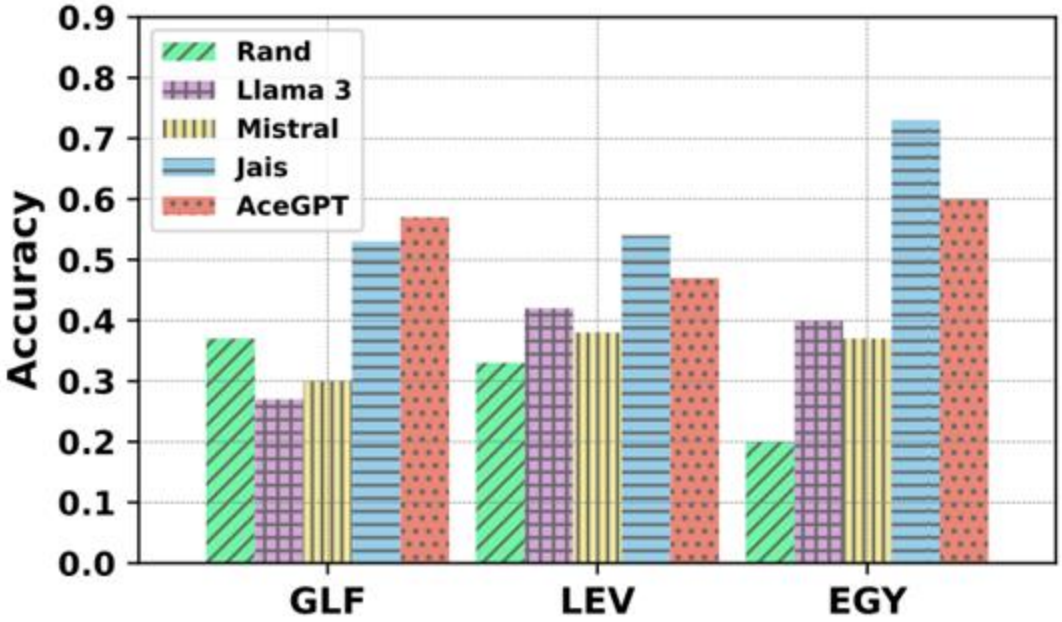
- Arabic-centric LLMs, such as Jais and AceGPT are better equipped to handle both MSA and its dialects
- Llama3 and Mistral, show a substantial deficiency in adapting to Arabic dialects, highlighting a significant gap in their ability to handle linguistic diversity in this context.

Cultural Understanding

- Arabic-centric models also demonstrate a superior understanding of cultural nuances compared to general multilingual LLMs.

AraDICE: Benchmarks for Dialectal and Cultural Capabilities in LLMs (Mousi et al., 2024)

AraDICE



AraDICE: Benchmarks for Dialectal and Cultural Capabilities in LLMs (Mousi et al., 2024)

AraDICE

AraDICE: Benchmarks for Dialectal and Cultural Capabilities in LLMs

https://huggingface.co/collections/QCRI/aradice-6727765839bf89aa78e9f132
https://arxiv.org/pdf/2409.11404

Today's Session

- Goals (5 minutes) 10.30 to 10.35
- Recent Work
  - ~~Native, Local and Cultural Aspects (Firoj) 11:25 - 11:35 in-person~~
  - LLMs on Everyday Knowledge in Diverse Cultures and Languages (Alice Oh) 10:35 - 10:50 in-person
  - CulturalVQA and Benchmarking VLMs for Cultural Understanding (Shravan) 10:50 - 11:15 in-person
  - Survey of Cultural Awareness in Language Models: Text and Beyond (Isabelle/Siddhesh) 11:15 - 11:25 in-person/Online
- Panel + Ask Questions 11:35 - 11:55
- Summary and Wrap-Up (5 minutes)

Panel + Open Discussion

Questions:

- LLMs struggle at capturing cultural nuances. How can we design LLMs to address cultural biases?
- The current approach to LLM training for non-English language is to use translated datasets. How challenging is this for multimodality? Moving forward, is it an ideal approach?

Summary and Wrap-Up

Slack channel